E350

U1300 and U3400

RELEASE NOTES FOR F/W VERSION 6.0

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Revision history

Version	Date	Comments
1	14/05/2019	Initial Release

Important Information
This document provides information relating to E350 U1300 Series 1, Series 2 and U3400 meter firmware (the "Product")

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1 About this document

1.1 Purpose & Scope

The purpose of this ***6.0 Firmware Release Note*** is to communicate new features, limitations and changes in meter functionality.

The 6.0 firmware release is for the E350 - U1300 and U3400 meters.

1.2 Definitions, Acronyms, and Abbreviations

RSM:	Relay State Machine for the load control relay
LC:	Load Control
LP:	Load Profile
CLC/R:	Controlled Load Contactor or Relay
CSM:	Customer Supply Monitoring including Neutral Integrity
QoS:	Quality of Supply
AMI:	Advanced Metering Infrastructure
AD:	Auto Disconnect
S2:	Series 2
SCC:	Supply Capacity Control
STS:	Smart Time Synchronisation
HES:	Head End Systems
E350:	Family name of Landis+Gyr modular meter including the types U1200, U1300, U1300 S2 U3300, U3400 and U3350
E650:	Family name of Landis+Gyr commercial and industrial meter including the type EM5400 and EM5400 S2
E420:	Family name of Landis+Gyr integrated meter S1100 RF and S1100 LTE
R1100:	Landis+Gyr’s smart load control device.

1.3 References

E350 U1300 User Manual

E350 U3400 User Manual

EMPWin+ User Manual

2 About the release, compatible products and Summary

This firmware package provides upgrade to the following meter/firmware:

Previously Released firmware version: R5.4

This Released Version: R6.00

U3400 Build Version: 016 (U3400) and 017 (U1300)

Compatible Meter	Firmware Upgrade*	Firmware Downgrade*
U1300	4.2 → 5.4 (Series 1 Only) ↓ 4.3 → 5.4 (Series 1 Only) ↓ 4.4 → 5.4 ↓ 4.5 → 5.4 ↓ 4.6 → 5.4 ↓ 4.8 → 6.0 ↓ 4.9 → 6.0 ↓ 5.0 → 6.0 ↓ 5.4 → 6.0	5.4 → 5.0
U3400	4.4 → 5.4 ↓ 4.5 → 5.4 ↓ 4.6 → 5.4 ↓ 4.7 → 5.4 ↓ 4.9 → 6.0 ↓ 5.0 → 6.0 ↓ 5.4 → 6.0	5.4 → 5.0

Important Note: *customer will be required to refer to the release notes for the particular firmware version to understand the limitations of the upgrade/downgrade. Please also refer to the intermediary firmware release notes when skipping meter firmware version.

This firmware has been tested on the following platforms:

- E350 U1300 meters (this includes Series 1 and Series 2)
- E350 U3400 meters
- EMPWin+ 3.5.15

The FT100 release 5.3 or earlier must not be used with 6.0 meter firmware.

EMPWin+ 3.5.15 or greater must be used with 6.0 meter firmware.

E350 U1300 S2 meters cannot be downgraded to any previous R4.3 or earlier F/W Version.

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2.1 New Features for Supported Meters

2.1.1 General New Features

The E350 U1300 & U3400 meters with 6.0 firmware and have the following enhancements.

- The F/W now supports the larger 64Mb memory chips and continues to support the 16Mb hardware variants of the U1300 and U3400 meters. The increased memory size can be used for load profiling of shorter interval lengths for much longer time periods.

- When using 6.0 F/W on the smaller 16Mb memory hardware variants, the meter program must have Extended 1 selected before writing this program to the meter.

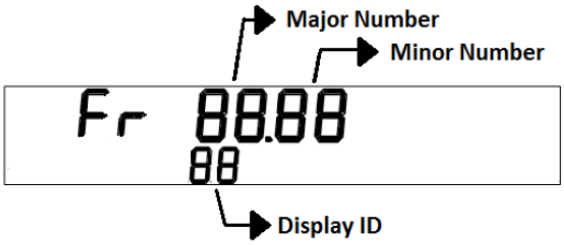
- When using 6.0 F/W on 64Mb memory hardware variants, the meter program must have Extended 2 selected before writing this program to the meter.

- MeterComm will also ensure only compatible programs are written to the meter.

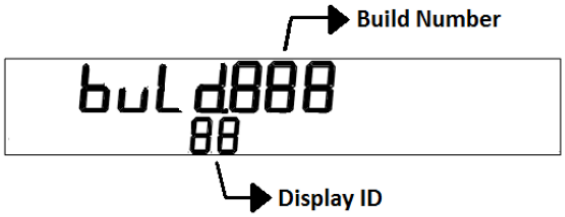
- The EMPWin+ LP file format has been changed to use JSON format that has benefits over the older xml format when processing large files.

- This firmware release 06.00.016 implements a new method of F/W identification that now utilises three separate fields; F/W Major Version 2 digits, Minor Version 2 digits and Build Version 3 digits.

- The display list now includes “Firmware Revision” and Firmware Build Number” as separate display items.



Major and Minor Number



Build Number

- The main differences between the two different memory size meters are tabulated below.

Feature	Extended 1 Memory (16Mb memory variants)	Extended 2 Memory (64Mb memory variants)
Daily Self Reads	35	200
TOU Registers	6	24
5 Min LP	Yes	Yes
LP Channels/Group	4	24

- **200 Daily Self Reads**
  The self-read is a snapshot of the accumulated energy totals and TOU registers at midnight each day. The meter stores the last 8 self reads when operating FW R4.9 or earlier. The meter stores the last 35 self reads when operating FW R5.0, R5.4 and 200 self reads when operating FW 6.0.

- **24 TOU registers**
  The U1300 and U3400 meters variants with the larger 64Mb memory now support 24 TOU registers. The additional TOU registers can be used to capture midnight snapshots for validation of LP energy channels. When using 6.0 F/W on smaller 16Mb memory hardware variants, Extended 1 memory setting is required and continues to support up to 6 registers.

- **5 Minute Load Profile Intervals.**
  The U1300 and U3400 meters support 5 minute intervals when running R5.0. R5.4 and 6.0 firmware.

- **24 Load Profile channels per group.**
  The U1300 and U3400 meters with the larger memory and 6.0 F/W now support 24 channels per group with a total of 96 channels across 4 groups.

2.1.2 Firmware Upgrade

- The E350 U1300 & U3400 meters with 16Mb memory can be F/W upgraded to 6.0.
- The LP data is maintained when the firmware upgrade process expands the available LP memory. However, a reprogram is required to utilise this memory.
- The E350 U1300 & U3400 meters will have a non-volatile error 0x08 after F/W upgrade as a result of the significant changes to the LP configuration and increased LP memory size. This should be used as a reminder to fully program the meter with the Extended 1 memory mcd file.
- F/W 6.0 is an Upgrade Gate, where all previous F/W versions must be upgraded to 6.0 before being upgrading to any future firmware version. The U1300 and U3400

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meters must and be fully programmed with EMPWin+ 3.5.15 via MeterComm or MCD Export and HES after the upgrade.
- U1300 and U3400 meters with 64Mb memory will be manufactured with 6.0 firmware.

2.1.3 Firmware Downgrade

Firmware downgrade is not supported.

2.1.4 Billing Self Read Periods

- E350 U1300 & U3400 meters that are upgraded to 6.0 F/W will automatically start recording 35 self-read periods instead of 8 as occurs in earlier F/W versions.
- If the E350 U1300 & U3400 meters are fitted with 64Mb memory (Extended 2), then the meter will store up to 200 self-read periods.

2.1.5 Load Profile

- The E350 U1300 & U3400 with R5.0, R5.4 and 6.0 firmware supports 5 minute Load profile interval selection and increased memory referred to as “Extended 1” and “Extended 2”. The “Extended 1” memory can store up to 220 days of 5 minute LP intervals for a single LP group with four channels enabled. The “Extended 2” memory can store up to 2,196 days of 5 minute LP intervals for a single LP group with four channels enabled.
- Meter configuration (MCD) files with 5 minute intervals enabled are restricted in MeterComm and MCDExport from being written to meters with older firmware.

2.2 Itron MV90, MVRS, IEE Support

MV90 has been tested with 6.0 F/W

- MV90 reads the whole 200 self-read records but will only report the most recent 8.
- MV90 can support the reading of 5 minute LP data.
- HHF files created by MeterViewer can be imported into MV90 with 5 minute LP data.
- MeterViewer HHF format can export up to the first 16 LP channels in a single LP group.

2.3 Command Center

2.3.1 Command Center Software Quality Assurance testing.

Command Center (CC) version 7.3.2.11 has been tested with 6.0 F/W.

- 6.0 F/W can be sent Over the Air (OTA) using CC.
- The U3400 and U1300 meters must be fully programmed OTA after upgrading to 6.0.
- After reprogramming the U3400 and U1300 will continue to push LP, Billing Reads and Events to CC.
- Cellular comms F/W 14.19.001 can be OTA upgraded to 14.21.001.

- CC can read LP data from meters that have 5, 10, 15, 30 or 60 minute LP intervals (maximum of four LP groups per meter).
- CC 7.3 can only support meters that have 1 to 4 LP channels per LP group.

2.4 Bug fixes in this release

Item No.	*Description*
TFS-11151	U1300 Receiving Ripple Telegrams with full ripple log when power outage occurs, can cause a self-check error and incorrect LP zero fill intervals. This is rare but possible. This issue has now been resolved in firmware 6.0.
TFS-11158 **TFS-11416**	It was identified with F/W R4.6 or greater, under certain conditions, the Ripple LC module could release control at midnight. It was also noticed placing a LC switch entry at midnight resolves this issue. The issue could occur when using one season without a midnight switch entry in the Ripple program and Failsafe is activated to use the time table scheduler. FW 6.0 has now resolved this issue and placing a LC switch entry at midnight is not required.
TFS-11410	It was discovered if a later date was set first in the Load Control Time Table, the desired action will not be executed. This issue has now been resolved with EMPWin+ 3.5.15. Load Control Time Table entries are now chronologically written to meter during programming.
TFS-2375	If the 10th breaker disconnection due to Emergency Supply Capacity Control (ESCC) occurs at midnight then when ESCC timed out and try to re-connect breaker, it did not re-close breaker automatically, this has now been fixed.
TFS-2246 **TFS-2245** **TFS-6268**	If the U1300 and U3400 meter with SSN comms module and running R5.0 and R5.4 firmware with a backwards time reset over the interval boundary is actioned, only the start time event is recorded in the log. This issue has now been resolved in firmware 6.0.
TFS-1410 **Mantis-3254**	It was noticed a Season Change event 24 on a U1300 can be logged multiple times in the event log during a partial programming operation. This issue has now been resolved in firmware 6.0.

2.5 Known Issues in this release

Item No.	*Description*
Mantis-3163	If there is a long power outage across a Season change date, then the Billing report might incorrectly label the season number as 0. The date logged, and energy/demand values are correct. This is an existing bug from previous firmware versions.
Mantis-3584	If there is a power outage across a LC season change from LC season 1 to LC season 0 then the event that indicates this LC season change to 0 is not recorded. The logging works correctly for LC season change from 0 to 1 across a power outage. The LC scheduler works correctly it is only the event that is missing.

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Mantis-3906	UFLC module remains active after power cycle and Module Nominal Condition has returned. Module nominal condition returned happens when the meter is powered down, therefore no event is recorded, however, if the outage duration is short about 3 seconds, UFLC module remains active on power up. An outage of 6 seconds correctly released the UFLC allowing the next highest RSM module to take control e.g. Ripple.
TFS-48967 **Mantis-3096**	Demand Limit Load Control If Under Frequency Load Control (UFLC) module is triggered due to bad frequency, then the frequency recovers and UFLC starts its Nominal Delay period AND then a short power cycle occurs (about 3 seconds) AND on Power Up the Frequency is still within Normal limits; under these conditions the UFLC module might not correctly exit its Nominal waiting period. This can cause the UFLC module to block other lower priority RSM commands such as Ripple telegrams. It appears that when this incorrect state is reached that it might require two power cycles to correctly reset the RSM logic.
TFS-43102 **TFS-1567** **Mantis-3904**	If the Ripple Power Restore option has been set to Time Table AND the random power restore range has been set to 0 then the Time Table schedule will not run. This is unlikely to occur because the random power restore is not likely to be set to 0 by customers.
TFS-2519	Corruption of Previous Billing data when firmware upgrading to 6.0 Meters can be F/W upgraded from R4.8, R4.9, R5.0, R5.4 to R6.0. The TOU data and LP data is maintained when the firmware upgrade process expands the TOU area from 8 to 35 days of self-reads. Previous Billing data can be corrupted when F/W upgraded to R6.0. In most applications the Previous Billing data is not used by HES systems or by customers. Re-programming the meter is required to enable the new functionality provided in 6.0 and will also clear the Previous Billing data.
TFS-48970 **TFS-1565** **Mantis-3918**	It has been observed that U1300 meters, the LC1 & LC2 random delay closed values tend to have the same minutes and seconds with only the hours being different. Random delay is not random enough for the two relays.
TFS-1566 **TFS-48701** **Mantis-3919**	It was noticed the Ripple Control (RC) icon on the LCD may not display correctly under the following scenario. The RC icon on the display should be slowly flashing if the meter contains a valid ripple program and it is in Failsafe mode. Writing meter program with Ripple support "False" settings to meter while it is in Failsafe mode should automatically cancel any ripple operation and set RC icon to permanent On state. It was noticed the RC icon was not set to permanent On state and continued flashing slowly.
TFS-1067 **TFS-43106**	With firmware versions 6.0 and earlier, It is possible that after installation the meter might not switch the LC relay when the first Ripple Telegram is received. If the meter has been programmed with Ripple, and a power outage occurs before the first valid telegram is sent, then the LC relay might not switch. Subsequent ripple telegrams are handled correctly.